

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Four Valve Spring Compressor Kit, Part Number 4919358, or equivalent
- Valve stem seal installer, Part Number 5296545, or equivalent
- Boot pliers, Part Number 3163293, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

Off-Engine Repair

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the cylinder head and set it on a bench or table. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/377/377-002-004.html>)

On-Engine Repair

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the exhaust gas recirculation (EGR) crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Disconnect the engine brake solenoids from the internal actuator wiring harness. Refer to Procedure 019-063 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-063.html>)
- Remove the rocker levers from the cylinder to be repaired. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-009.html>)
- Remove the injector from the cylinder to be repaired. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/431/431-006-026.html>)
- Bar the engine over so the cylinder is at top dead center (TDC). Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-004-tr.html>) Refer to Procedure 000-017 in Section 0. (</qs3/pubsys2/xml/en/procedures/377/377-000-017.html>)

Repeat this step as needed for each cylinder.

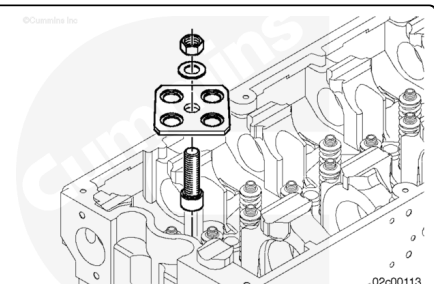
Remove

⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.

⚠ CAUTION ⚠

**Do not use an impact tool for any part of this procedure.
Doing so can damage the tool and/or the engine.**

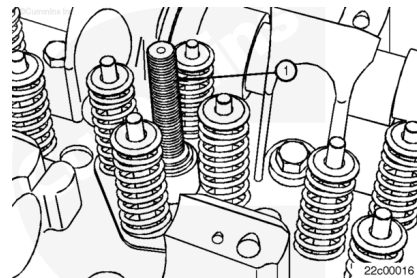


Note : This step can be performed either on- or off-engine.

Use four valve spring compressor kit, Part Number 4919358, or equivalent, to compress the valve springs.

Remove the spring retainer collets.

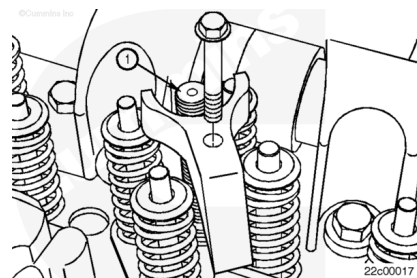
Position the replacer screw (1) in the injector hole with the threaded end pointing up.



Secure the replacer screw (1) in the injector hole with the injector clamp and capscrew.

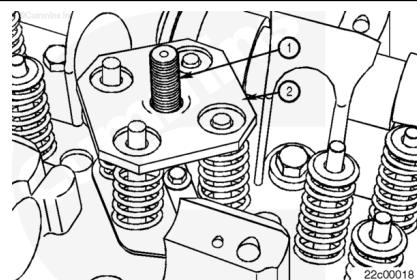
Tighten the capscrew.

Torque Value: 54 n•m [40 ft-lb]



⚠ CAUTION ⚠

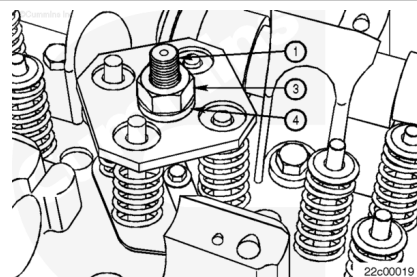
The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs. Failure to position the retainer plate properly can result in damage to the engine and/or tool.



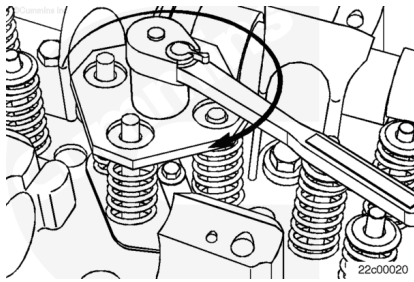
Position the compressor plate (2) over the replacer screw (1) and onto the valve springs.

Install the washer (4) and forcing nut (3) on the replacer screw (1).

Tighten hand-tight.



Turn the forcing nut (3) **clockwise** until the valve springs start to compress.

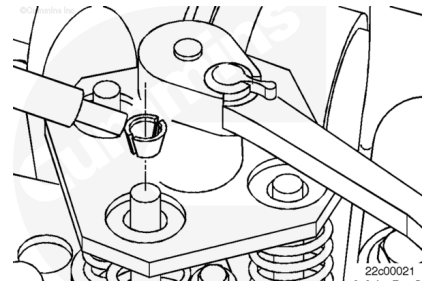


⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.

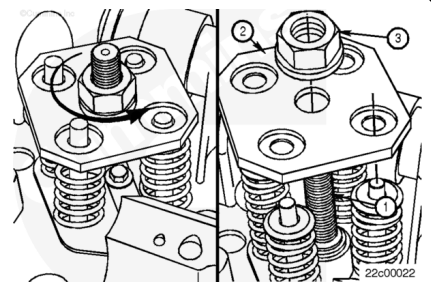
Continue turning the forcing nut (3) **clockwise** until the valve collets can be removed with a magnetic tool.

Discard the valve collets.

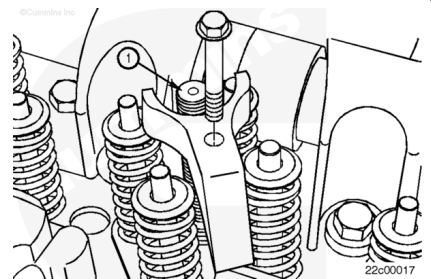


Carefully turn the forcing nut (3) **counterclockwise** until the tension is released from the valve springs.

Remove the forcing nut (3) and compressor plate (2) from the replacer screw (1).

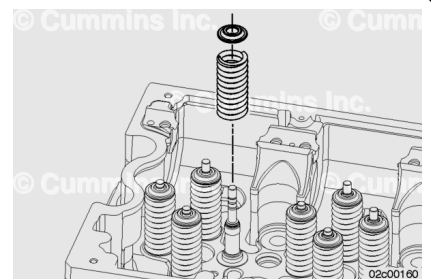


Remove the replacer screw (1) by removing the injector hold-down bolt and the injector hold-down clamp.



Remove the following:

- Valve spring retainer
- Valve spring
- Valve spring guide.

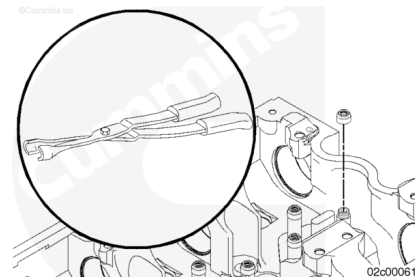


Note : Both valve stem seals and valve guide seals can be installed on an engine. Valve stem seals are fitted over the valve guide and have an integrated valve spring. Valve guide seals are fitted to the valve guide.

For valve stem seals with an integrated valve spring guide, remove the part by hand.

For valve guide seals, remove the part using boot pliers, Part Number 3163293, or equivalent.

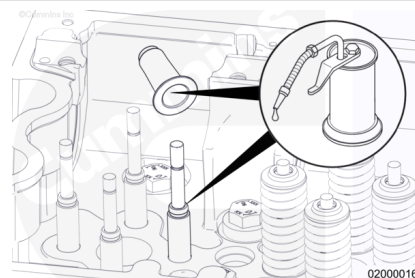
Repeat the above steps for each cylinder as needed.



Install

Lubricate the top of the valve guide and the inside diameter of the new valve stem seal with clean engine oil.

Note : Valve stem seals are **not** reusable and **must** be discarded if removed for any reason.



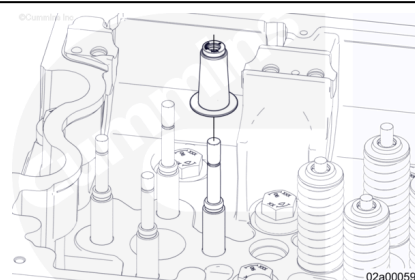
⚠ CAUTION ⚠

To avoid engine and/or tool damage, do not use a service tool to install valve stem seals with an integrated valve spring guide.

⚠ CAUTION ⚠

To avoid engine damage, make sure the valve spring guides are removed before attempting to install a valve stem seal.

Locate the valve stem seal over the valve guide and firmly push the seal into place by hand until the integrated valve spring guide is seated firmly against the cylinder head.

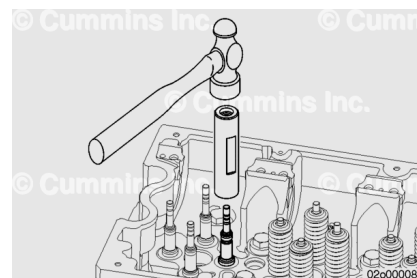


⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.

Use the valve stem seal installer, Part Number 5296545, or equivalent, to seat the valve guide seals on the valve guides.

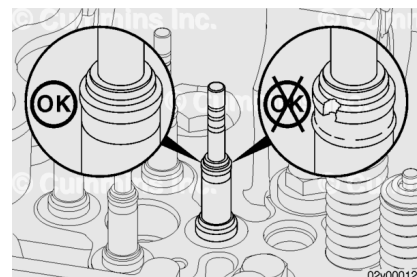
Once the tool contacts the valve spring guide, the seal is fully installed.



Inspect the valve stem or guide seal for any damage after installation.

If damage is found, the seal **must** be removed and a new seal installed.

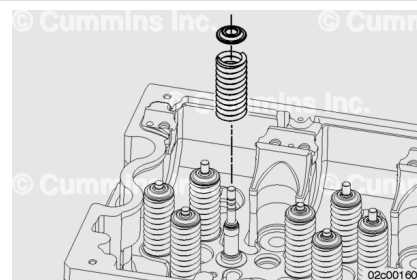
Note : For valve guide seals, a gap of up to 2 mm [0.08 in] between the valve guide and the bottom of the seal is acceptable.



Install the following:

- Valve spring
- Valve spring retainer.

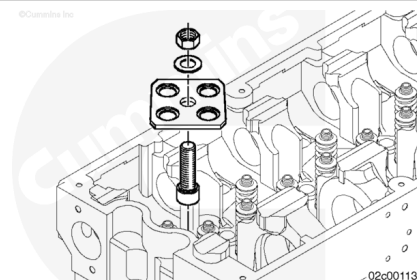
Repeat the above steps for each cylinder as needed.



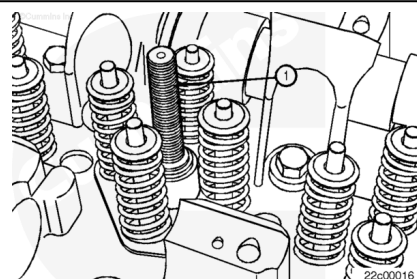
⚠ CAUTION ⚠

**Do not use an impact tool for any part of this procedure.
Doing so can damage the tool and/or the engine.**

Use the four valve spring compressor kit, Part Number 4919358, or equivalent, to compress the valve springs.



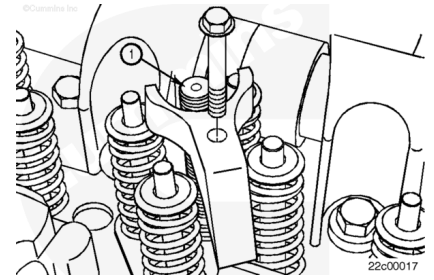
Position the replacer screw (1) in the injector hole with the threaded end pointing up.



Secure the replacer screw (1) in the injector hole with the injector clamp and capscrew.

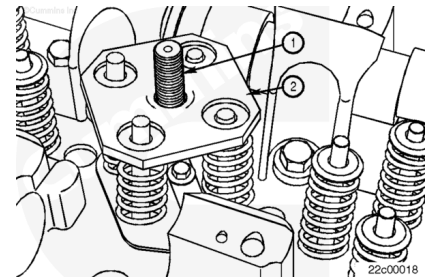
Tighten the capscrew.

Torque Value: 54 n•m [40 ft-lb]



⚠ CAUTION ⚠

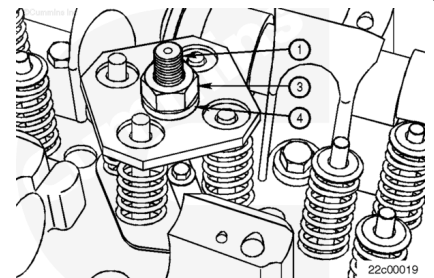
The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs. Failure to position the retainer plate properly can result in damage to the engine and/or tool.



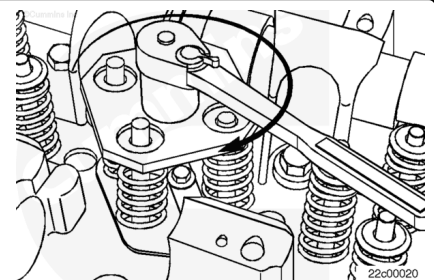
Position the compressor plate (2) over the replacer screw (1) and onto the valve springs.

Install the washer (4) and forcing nut (3) on the replacer screw (1).

Tighten the nut hand-tight.



Turn the forcing nut (3) **clockwise** until the valve springs start to compress.

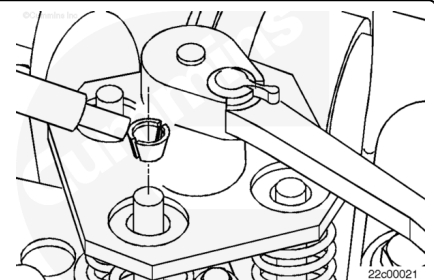


⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.

Continue turning the forcing nut (3) **clockwise** until the valve collets can be installed.

Always use new collets when reassembling the valve spring retainers.

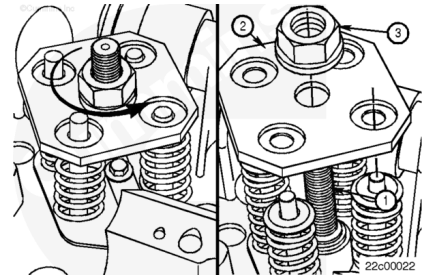


Install the new valve spring retainer collets.

Inspect the collets for proper seating and installation.

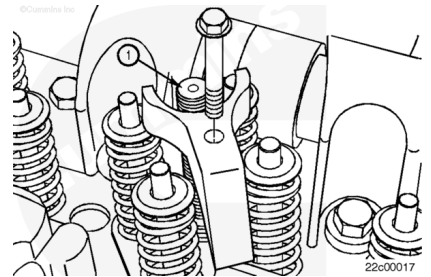
Carefully turn the forcing nut (3) **counterclockwise** until the tension is released from the valve springs.

Remove the forcing nut (3) and compressor plate (2) from the replacer screw (1).



Remove the replacer screw (1) by removing the injector hold-down bolt and the injector hold-down clamp.

Repeat the above steps for each cylinder as needed.



Finishing Steps

On-Engine Repair

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-026.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-009.html>)
- Connect the engine brake solenoids to the internal actuator wiring harness. Refer to Procedure 019-063 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-063.html>)

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html>)
- If damage resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.
- Use the following procedure to check the aftertreatment selective catalytic reduction (SCR). Refer to Procedure 014-025 in Section 14. (</qs3/pubsys2/xml/en/procedures/377/377-014-025.html>)
- Use the following procedure to check the aftertreatment diesel particulate filter (DPF). Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.

Off-Engine Repair

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/377/377-002-004.html>)
- If damage resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.
- Use the following procedure to check the aftertreatment SCR. Refer to Procedure 014-025 in Section 14. (</qs3/pubsys2/xml/en/procedures/377/377-014-025.html>)
- Use the following procedure to check the aftertreatment DPF. Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)
- Operate the engine to normal operating temperature. Check for leaks and proper operation.